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CRYPTOGRAPHY

EXP NO: 5

SNORT

AIM: To Learn how to use Snort to detect real-time threats, analyse recorded traffic files and identify anomalies.

TASK 2 :

- Navigate to the Task-Exercises folder and run the command "./.easy.sh" and write the output
  → Too Easy!

TASK 3 :

- Which IDS or IPS type can help you stop the threats on a local machine?
  → HIPS
- Which IDS or IPS type can help you detect threats on a local network?
  → NIDS
- Which IDS or IPS type can help you detect the threats on a local machine?
  →HIDS
- Which IDS or IPS type can help you stop the threats on a local network?
  → NIPS
- Which described solution works by detecting anomalies in the network?
  → NBA
- According to the official description of the snort, what kind of NIPS is it?
  → full-blown
- NBA training period is also known as ...
  → baselining

TASK 4 :

- Run the Snort instance and check the build number.
  → 149
- Test the current instance with "/etc/snort/snort.conf" file and check how many rules are loaded with the current build.
  → 4151
- Test the current instance with "/etc/snort/snortv2.conf" file and check how many rules are loaded with the current build.
  → 1

TASK 6 :

- Now, you should have the logs in the current directory. Navigate to folder "145.254.160.237". What is the source port used to connect port 53?
  → 3009
- Read the snort.log file with Snort; what is the IP ID of the 10th packet?
  → 49313
- Read the "**snort.log.1640048004"** file with Snort; what is the referer of the 4th packet?
  → http://www.ethereal.com/development.html
- Read the "**snort.log.1640048004"** file with Snort; what is the Ack number of the 8th packet?
  → 0x38AFFFF3
- Read the "**snort.log.1640048004"** file with Snort; what is the number of the **"TCP port 80"** packets?
  → 41

TASK 7 :

- What is the number of the detected HTTP GET methods?
  → 2

TASK 8 :

- What is the number of the generated alerts?
  → 170
- Keep reading the output. How many TCP Segments are Queued?
  → 18

- Keep reading the output.How many "HTTP response headers" were extracted?
  → 3
- What is the number of the generated alerts?
  → 68
- What is the number of the generated alerts?
  → 340
- Keep reading the output. What is the number of the detected TCP packets?
  → 82
- What is the number of the generated alerts?
  → 1020

TASK 9 :

- Use "task9.pcap". Write a rule to filter IP ID "35369" and run it against the given pcap file. What is the request name of the detected packet? You may use this command: "snort -c local.rules -A full -l . -r task9.pcap"
  → TIMESTAMP REQUEST
- Clear the previous alert file and comment out the old rules. Create a rule to filter packets with **Syn** flag and run it against the given pcap file. What is the number of detected packets?
  → 1
- Clear the previous alert file and comment out the old rules. Write a rule to filter packets with **Push-Ack** flags and run it against the given pcap file. What is the number of detected packets?
  → 216
- Clear the previous alert file and comment out the old rules. Create a rule to filter **UDP** packets with the same source and destination IP and run it against the given pcap file. What is the number of packets that show the same source and destination address?
  → 7
- Case Example - An analyst modified an existing rule successfully. Which rule option must the analyst change after the implementation?
  → rev

Room completed: 100%

Which IDS or IPS type can help you stop the threats on a local machine?

HIPS ✓ Correct Answer

Which IDS or IPS type can help you detect threats on a local network?

NIDS ✓ Correct Answer

Which IDS or IPS type can help you detect the threats on a local machine?

HIDS ✓ Correct Answer

Which IDS or IPS type can help you stop the threats on a local network?

NIPS ✓ Correct Answer

Which described solution works by detecting anomalies in the network?

NBA ✓ Correct Answer

Answer the questions below

Run the Snort instance and check the build number.

149 ✓ Correct Answer

Test the current instance with "/etc/snort/snort.conf" file and check how many rules are loaded with the current build.

4151 ✓ Correct Answer

Test the current instance with "/etc/snort/snortv2.conf" file and check how many rules are loaded with the current build.

1 ✓ Correct Answer

3009 ✓ Correct Answer

Use **snort.log.1640048004**

Read the snort.log file with Snort; what is the IP ID of the 10th packet?

```
snort -r snort.log.1640048004 -n 10
```

49313 ✓ Correct Answer

Read the "**snort.log.1640048004**" file with Snort; what is the referer of the 4th packet?

http://www.ethereal.com/development.html ✓ Correct Answer

Read the "**snort.log.1640048004**" file with Snort; what is the Ack number of the 8th packet?

0x38AFFFF3 ✓ Correct Answer

Read the "**snort.log.1640048004**" file with Snort; what is the number of the **"TCP port 80"** packets?

Room completed: 100%

Investigate the **mx-1.pcap** file with the default configuration file.

```
sudo snort -c /etc/snort/snort.conf -A full -l . -r mx-1.pcap
```

What is the number of the generated alerts?

170 ✓ Correct Answer

Keep reading the output. How many TCP Segments are Queued?

18 ✓ Correct Answer

Keep reading the output.How many "HTTP response headers" were extracted?

3 ✓ Correct Answer

Investigate the **mx-1.pcap** file **with the second** configuration file.

```
sudo snort -c /etc/snort/snortv2.conf -A full -l . -r mx-1.pcap
```

What is the number of the generated alerts?

68 ✓ Correct Answer





RESULT: Thus we have learnt how to use Snort to detect real-time threats, analyse recorded traffic files and identify anomalies.